Nathan Au

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Education

Concordia University — **Montreal, QC**
Bachelor of Engineering – Software Engineering, Co-op — Sep. 2024 – May 2028 (expected)

- Program GPA: 3.72/4.00

Relevant Skills

Technical Languages: Python, TypeScript, JavaScript, SQL, Dart, HTML, CSS, C++, Java
Frameworks & Libraries: FastAPI, Flutter, React, Next.js, Pandas, Matplotlib, NumPy, Pytest, Hugging Face Transformers, OpenCV, Ultralytics YOLO, Google ADK, Pydantic, LangChain, LiveKit, Zod, Stripe, Twilio
Practices: OOP, REST APIs, Agile Scrum, Git Version Control, Unit & E2E Testing, ETL Data Pipelines, LLM Integration, Agentic Systems, Prompt Engineering, CI/CD, Database Design, Async Programming, System Design
Spoken Languages: English (Native), French (Professional – CECR DELF B2)

Experience

Software Engineering Intern — **May 2026 – Aug. 2026**
Dendrometric — Montreal, QC

- Incoming Summer 2026

Software Engineering Intern — **Jan. 2026 – Apr. 2026**
Biotics Labs — Montreal, QC

- Scaled a conversational Twilio SMS user onboarding pipeline that collects 30+ data points with async message processing for metrics calculation, context generation, and nutrition plan creation to eliminate production timeouts.
- Shipped end-to-end full-stack features from frontend UI (Next.js, shadcn/ui) to backend API design (FastAPI), database schema and authentication (Supabase), payment processing (Stripe) and production CI/CD (Railway).
- Architected a LiveKit-based agentic voice system with tool-calling LLMs, custom SessionManagers for persistent multi-turn conversations, and database-integrated tools to autonomously recommend and collect meal orders.
- Deployed production LLM infrastructure using Gemini and LangChain with Langfuse prompt management and context aggregation from user profiles and conversation history to enable personalized interactions at scale.

Software Developer — **Nov. 2024 – Apr. 2025**
Google Developer Group Montreal — Montreal, QC

- Developed a cross-platform Flutter mobile application that helps university students in Montreal connect with peers, discover local study spots, and stay up to date on upcoming campus events all on one platform.
- Built a data-driven recommendation engine using user profiles managed in Firebase Realtime Database and Gemini LLM integration via Vertex AI to generate personalized connection suggestions based on shared interests and goals.
- Integrated the Concordia University Open Data API to retrieve 60+ upcoming campus events in real-time and implemented responsive geolocation-based features using flutter_map for interactive study spot discovery.
- Delivered technical presentations at Flutter Montreal and Concordia University to audiences of 100+ attendees and produced the official launch video to showcase system architecture and data flow.

Projects

Reel Digest | Python, Telegram Bot API, Ollama (Granite 4), SQLite, yt-dlp, OpenCV | Source Code — **Dec. 2025**
Messaging-based productivity tool that saves users' time by turning short-form content into concise text summaries.

- Designed a multi-modal media processing pipeline with Google STT for audio transcription and OpenCV computer vision/Tesseract OCR optical character recognition for on-screen text extraction to enable contextual LLM analysis.
- Configured a SQLite caching layer to eliminate redundant inference, persist user history, and successfully serve live summaries for 300+ unique production requests.

TeeFour AI | Python, FastAPI, Ollama (Gemma 3), SQLModel, Tesseract OCR, Pytest | Source Code — **Oct. 2025**
Automated tax document processing system that eliminates manual data entry for tax accounting workflows.

- Engineered a two-layer document classification algorithm using PyMuPDF/Tesseract OCR to categorize known document types (T4s, IDs, receipts) via filename analysis and content-keyword matching logic.
- Streamlined tax form data field extraction by integrating a locally hosted LLM to transform unstructured OCR text inputs into validated, machine-readable JSON outputs for downstream use in tax filing software.

More projects: nathan-au.github.io